Benha University
Faculty of Engineering (Shoubra)
Electrical Engineering

ELE123 Technical Reports

ELE114 Technical Report Writing

(2026-2027)

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Graphical Representation of Data

Lecture 5

Data

- Data are individual facts, statistics, or items of information, often numeric.
- In a more technical sense, data are a set of values of qualitative or quantitative variables about one or more persons or objects, while a datum (singular of data) is a single value of a single variable.

Graphical Representation of Data

- We use graphics to understand the problem under investigation. Pictures can convey an overall message much better than a list of numbers.
- In this lecture, we describe some graphical representations of data.

1- Dot Plot

- Dot plot is a graphical representation of numerical data. A dot plot is a number line with x's placed above specific numbers to show their **frequency**. By the frequency of a number, we mean the number of occurrence of that number.

Example:

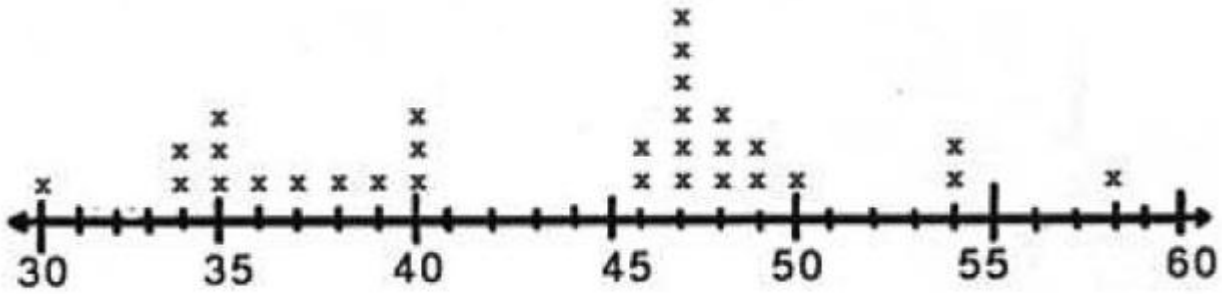
Suppose thirty people live in an apartment building with the following ages:

58 30 37 36 34 49 35 40 47 47

39 54 47 48 54 50 35 40 38 47

48 34 40 46 49 47 35 48 47 46

Make a dot plot of the ages.



- This graph shows all the ages of the people who live in the apartment building. It shows that the youngest person is 30, and the oldest is 58. Most people in the building are over 46 years of age. The most common age is 47.

- Dot plots allow several features of the data to become more obvious. For example, outliers, clusters, and gaps are realized.
  - **Outliers** are data points whose values are significantly larger or smaller than other values, such as the ages of 30, and 58.
  - **Clusters** are isolated groups of points, such as the ages of 46 through 50.
  - **Gaps** are large spaces between points, such as 41 and 45.

2- Histogram

- When we deal with **large sets of data**, a good overall picture and sufficient information can often be conveyed by distributing the data into several **classes** and determining the number of elements belonging to each class, called **class frequency**.

- For instance, the following table shows some test scores of a course.

65 91 85 76 85 87 79 93 82 75 100 70 88 78 83 59

87 69 89 54 74 89 83 80 94 67 77 92 82 70 94 84

96 98 46 70 90 96 88 72

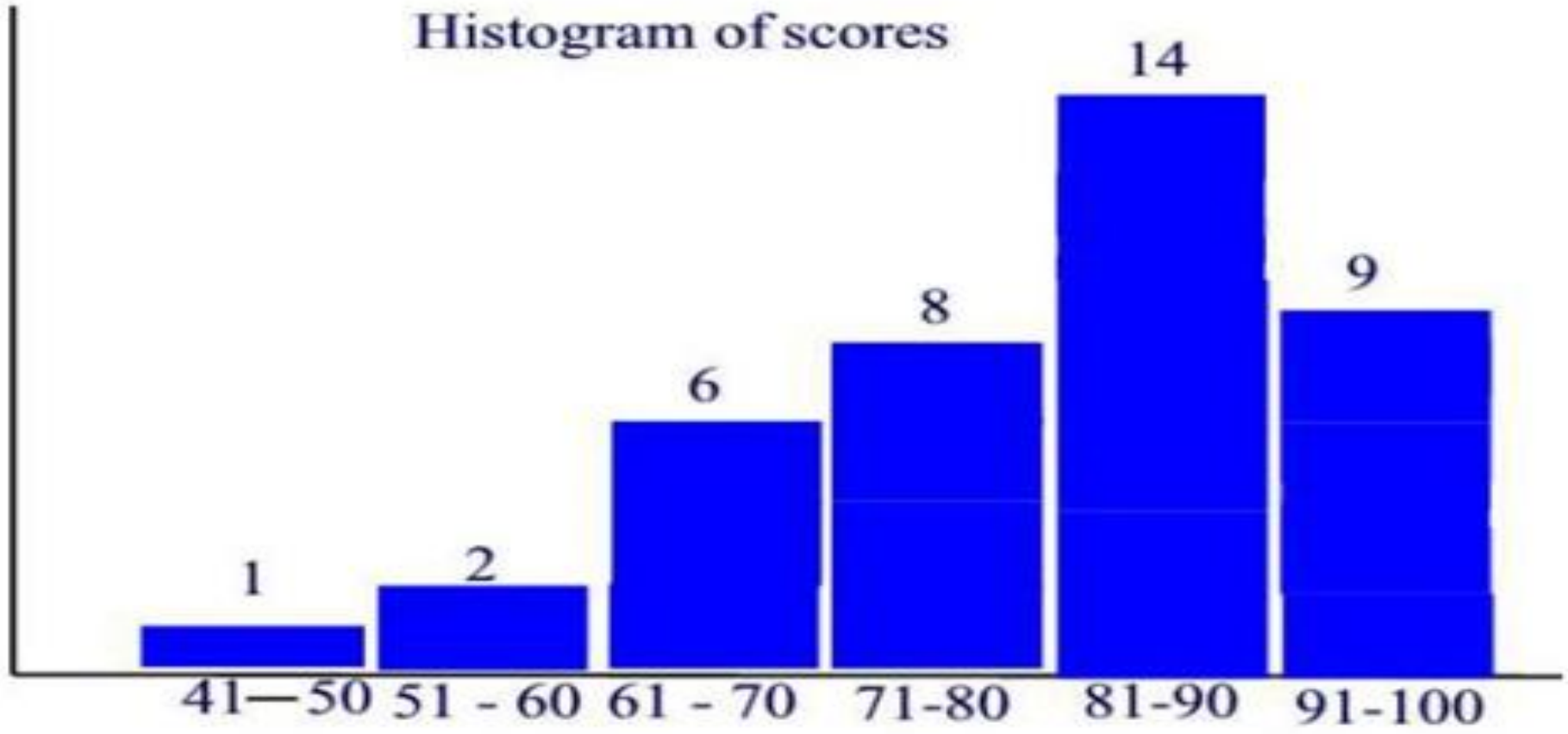
It's hard to get a feel for this data in this format because it is unorganized.
To construct a frequency distribution,

$$\text{Compute the class width } CW = \frac{\textit{Largest data value} - \textit{smallest data value}}{\textit{Desirable number of classes}}$$

- Round CW to the next highest whole number so that the classes cover the whole data.
- Thus, if we want to have 6 class intervals, then CW = (100−46)/6 = 9.
- The low number in each class is called the lower-class limit, and the high number is called the upper-class limit.
- With the above information we can construct the following table called **frequency distribution**.

Class	Frequency
41-50	1
51-60	2
61-70	6
71-80	8
81-90	14
91-100	9

- Once frequency distributions are constructed, it is usually advisable to present them graphically. The most common form of graphical representation is the histogram.

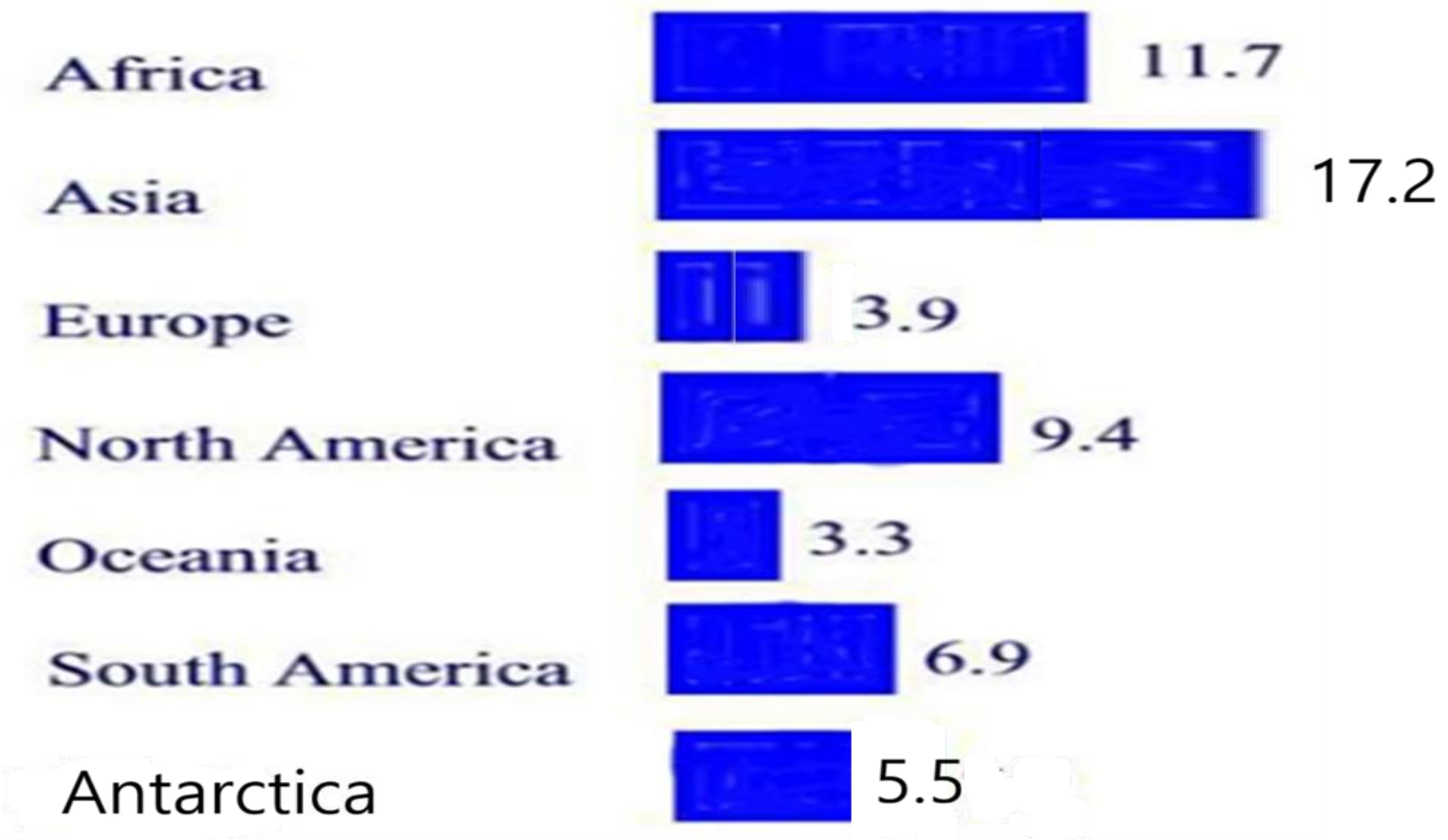


3- Bar Graph

- A bar graph is similar to a histogram.
- It is often useful in conveying information about **categorical data** where the horizontal scale represents a **non-numerical** attribute.
- The bars can be vertical or horizontal. Vertical Bar Charts are called Column Charts.
- The length of a bar represents the quantity we wish to compare.

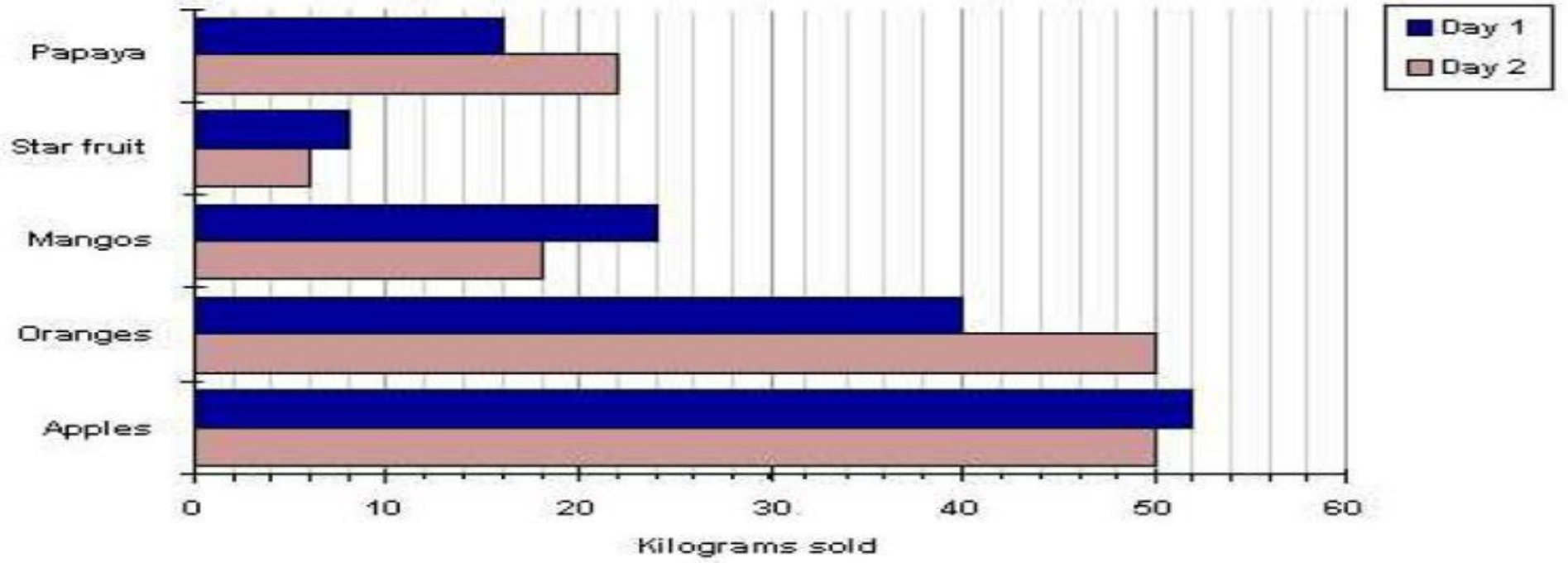
Example:

The areas of the various continents of the world (in millions of square miles) are: 11.7 for Africa; 17.2 for Asia; 3.9 for Europe; 9.4 for North America; 3.3 for Oceania; 6.9 for South America; 5.5 for Antarctica. Draw a bar chart representing the above data and where the bars are **horizontal**.



A **double bar graph** is similar to a regular bar graph, but gives 2 pieces of information for each item on the vertical axis, rather than just 1.

The bar chart given below, shows the weight in kilograms of some fruit sold on two different days by a local market. This lets us compare the sales of each fruit over a 2-day period, not just the sales of one fruit compared to another. We can see that the sales of star fruit and apples stayed most nearly the same. The sales of oranges increased from day 1 to day 2 by 10 kilograms. The same amount of apples and oranges was sold on the second day.

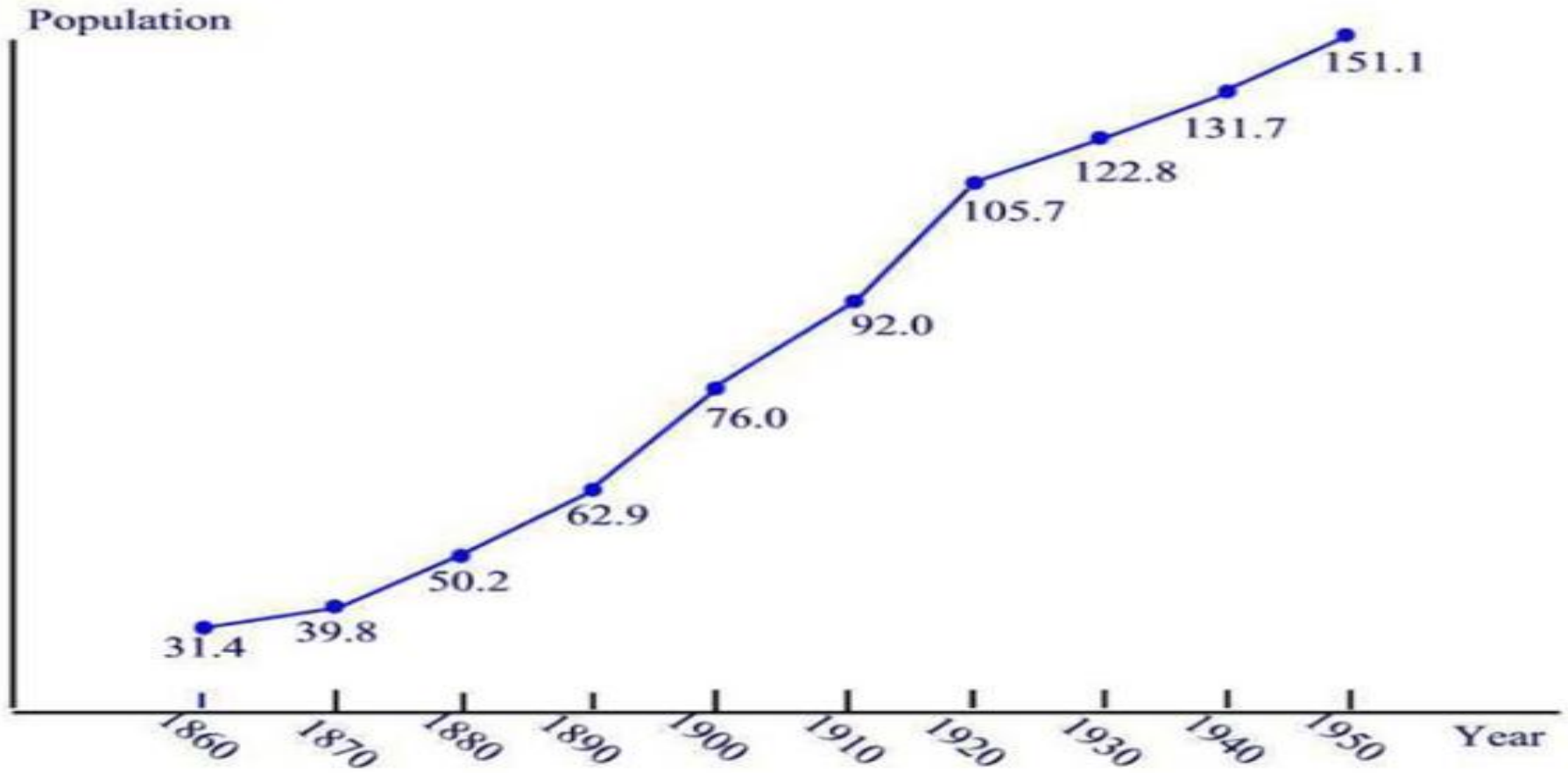


4- Line Graph

- A Line graph is particularly appropriate for representing data that vary continuously.
- A line graph typically shows the trend of a variable over time.
- To construct a line plot, we put time on the horizontal scale and the variable being measured on the vertical scale and then we connect the points using line segments.

Example:

The population (in millions) of the US for the years 1860-1950 is as follows: 31.4 in 1860; 39.8 in 1870; 50.2 in 1880; 62.9 in 1890; 76.0 in 1900; 92.0 in 1910; 105.7 in 1920; 122.8 in 1930; 131.7 in 1940; and 151.1 in 1950. Make a line plot showing this information.



5- Pie Chart

- A circle graph or pie chart, consists of a circular region partitioned into disjoint sections, with each section representing a part or percentage of a whole.
- To construct a pie chart, we first convert the distribution into a percentage distribution.
- Then, since a complete circle corresponds to 360 degrees, we obtain the central angles of the various sectors by multiplying the percentages by 3.6.
- We illustrate this method in the next example.

Example:

A survey of 1000 adults uncovered some interesting housekeeping secrets. When unexpected company comes, where do we hide the mess? The survey showed that 68% of the respondents toss their mess in the closet, 23% shove things under the bed, 6% put things in the bathtub, and 3% put the mess in the freezer. Make a circle graph to display this information.

We first find the central angle corresponding to each case:

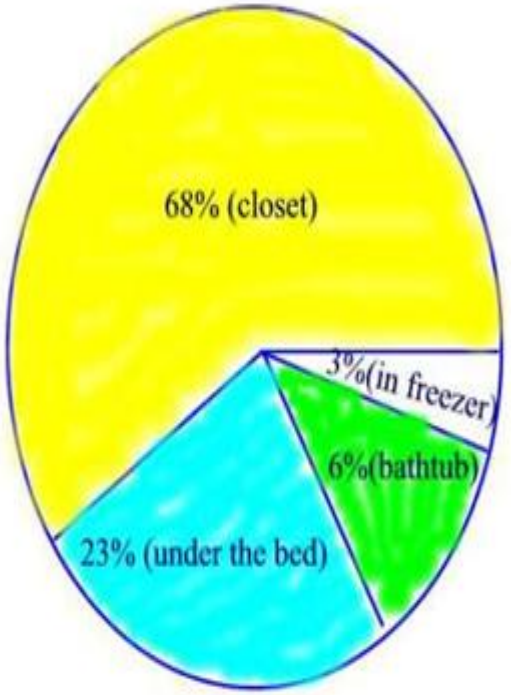
in closet $68 \times 3.6 = 244.8$

under bed $23 \times 3.6 = 82.8$

in bathtub $6 \times 3.6 = 21.6$

in freezer $3 \times 3.6 = 10.8$

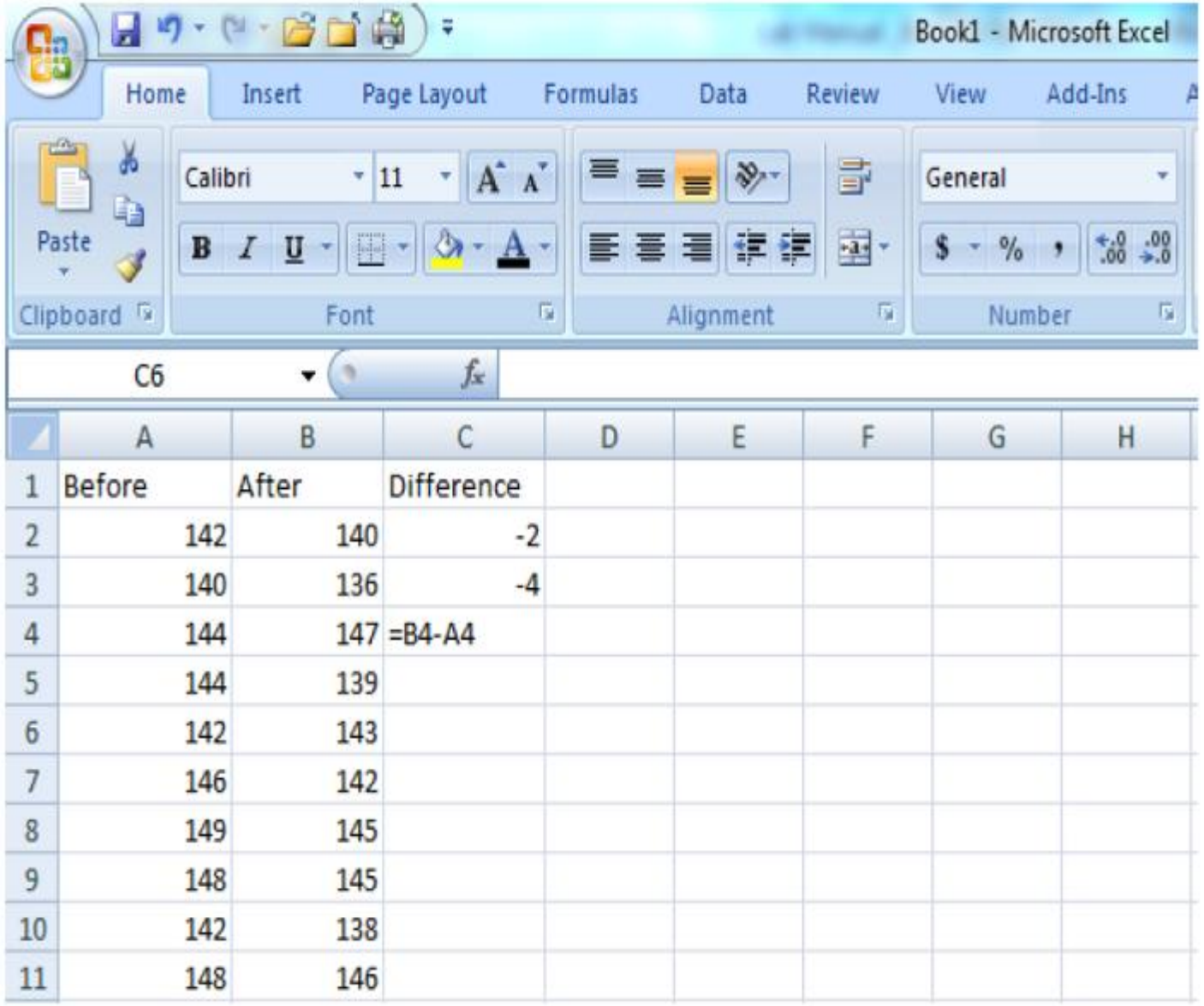
Note that $244.8 + 82.8 + 21.6 + 10.8 = 360$.



Data Analysis and Graphing in Excel

1- How to Enter Data in Excel

1. Open an excel worksheet
2. In the first row, label column A, column B, and column C
3. Enter raw data into the columns A and B
4. In the column C, enter a function (equation or formula starts with " = " symbol) to tell Excel to subtract the values in columns A and B, as shown in figure.

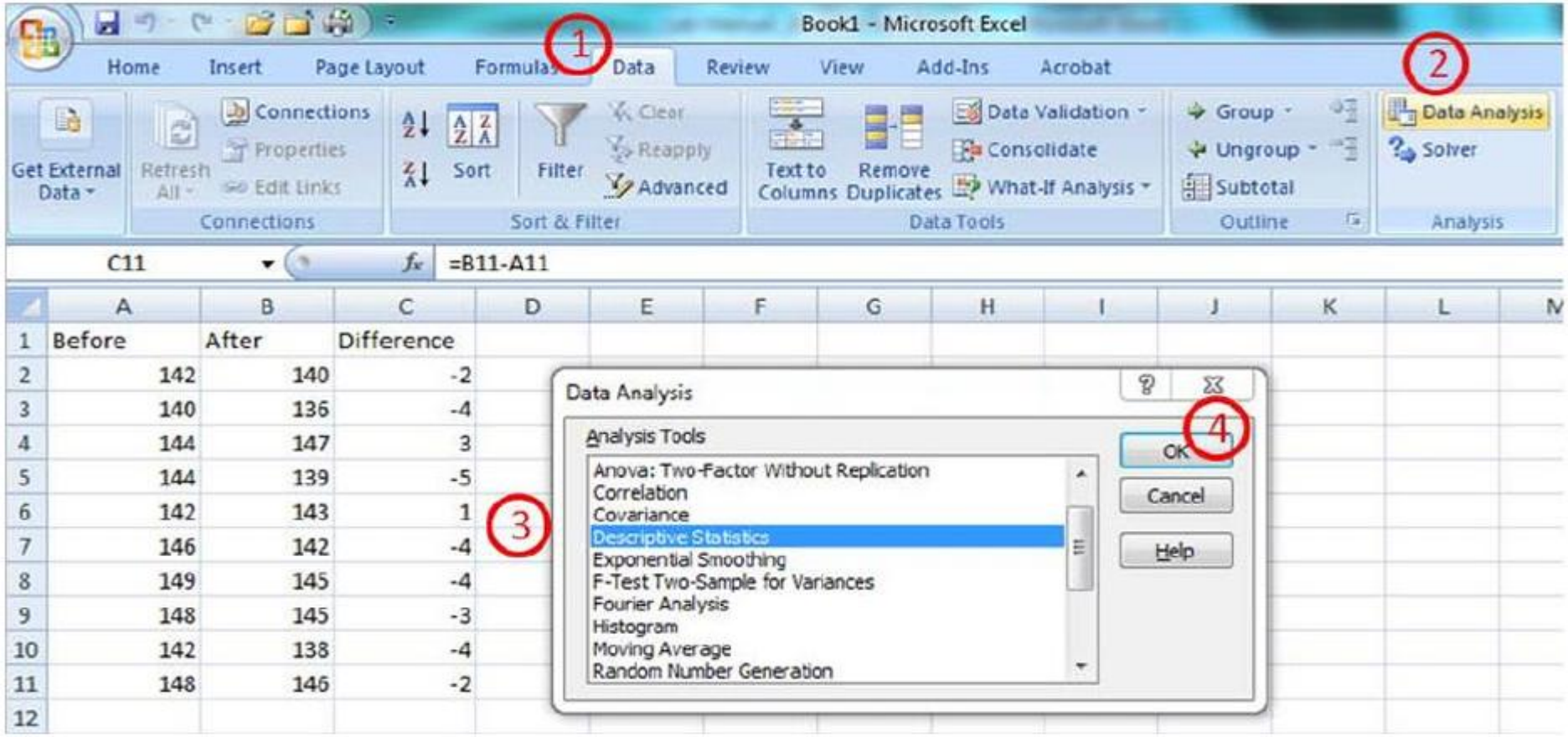


	A	B	C	D	E	F	G	H
1	Before	After	Difference					
2	142	140	-2					
3	140	136	-4					
4	144	147	=B4-A4					
5	144	139						
6	142	143						
7	146	142						
8	149	145						
9	148	145						
10	142	138						
11	148	146						

Data Analysis and Graphing in Excel

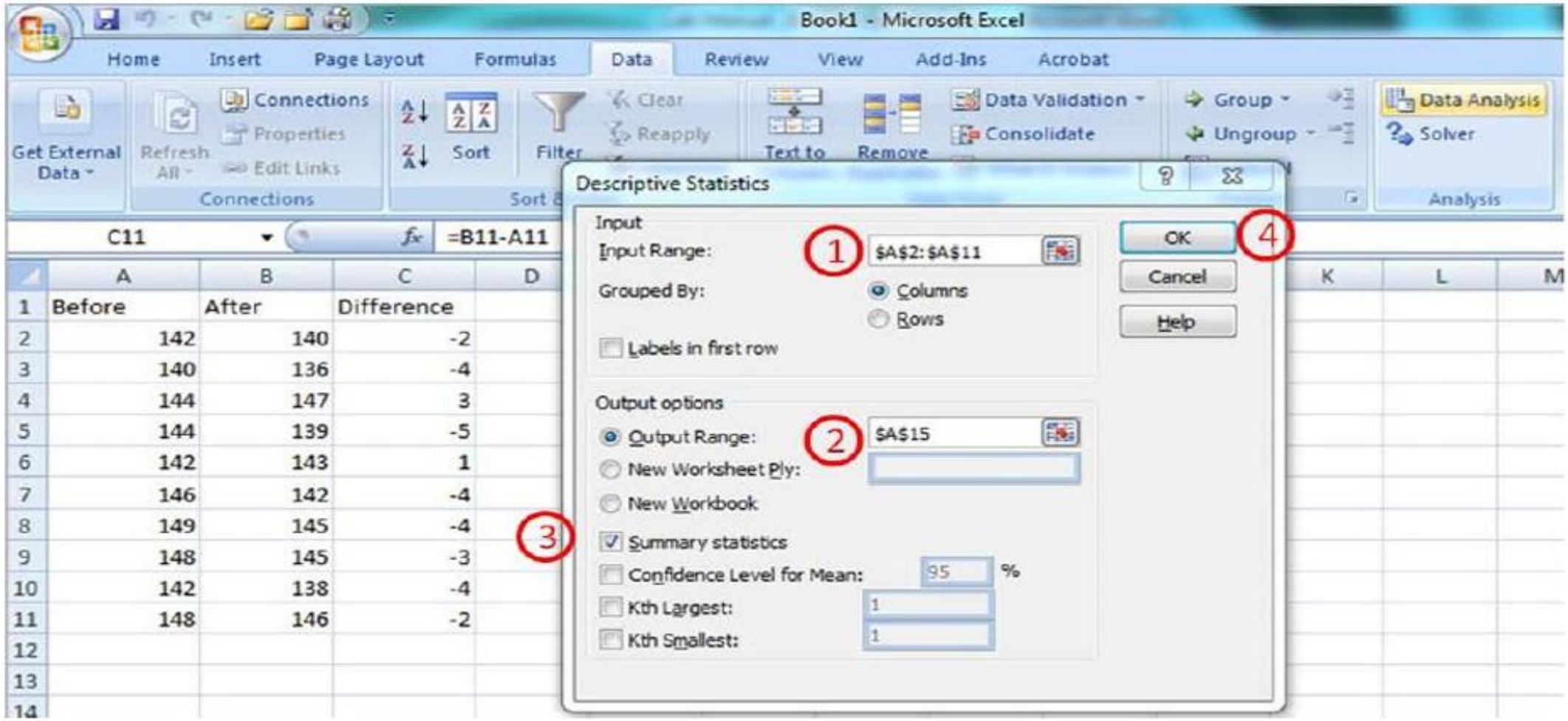
2- How to Calculate Summary Statistics in Excel

Open the worksheet with your data. Select the "Data" tab ①, then "Data Analysis" ②, then from the list choose "Descriptive Statistics" ③ and select "OK" ④.



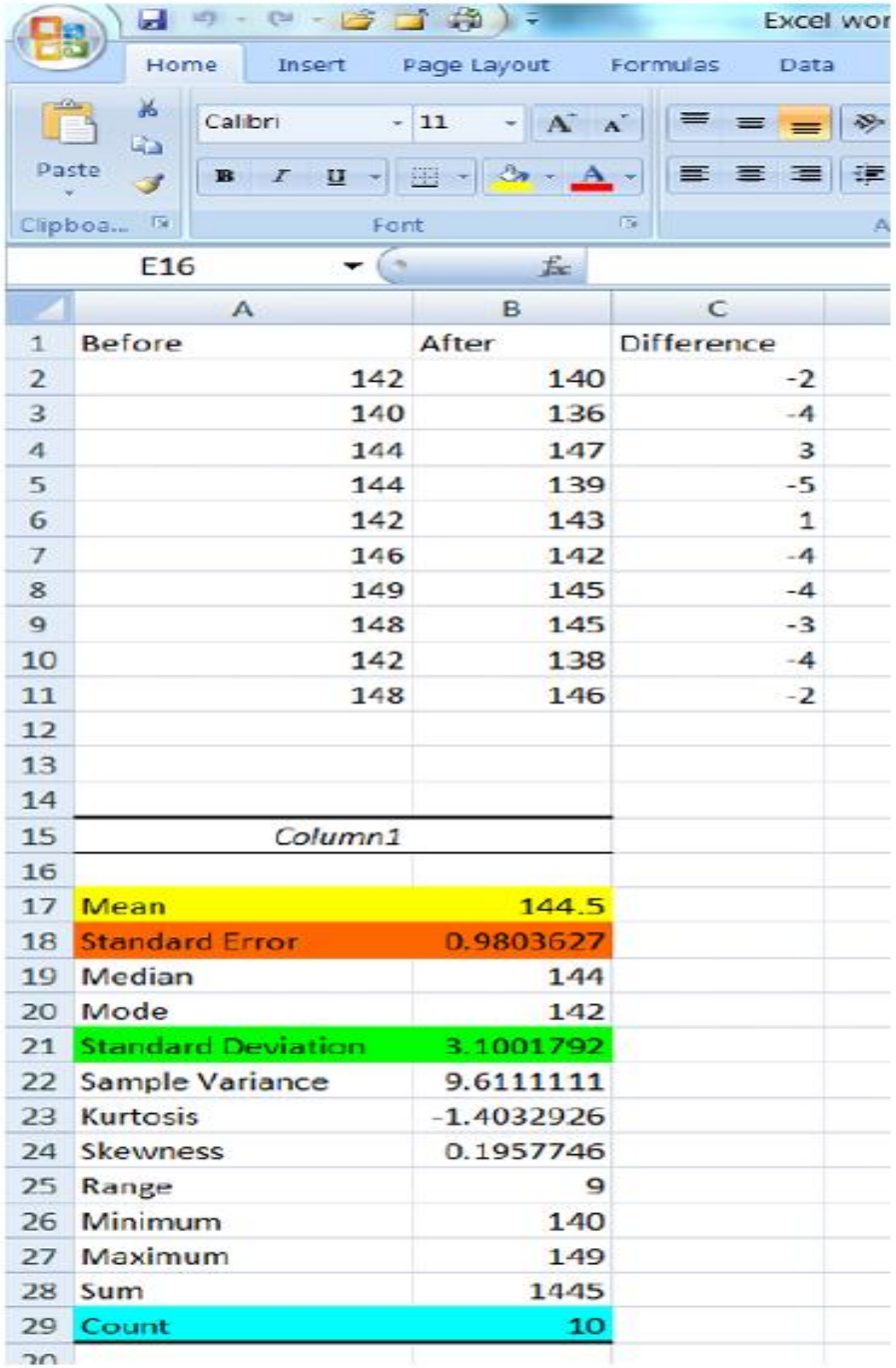
From here, with your cursor in the Input Range ①, select your data in column A.

Then put your cursor in the Output Range block ② and select an empty cell in work sheet. Click the box labeled "Summary Statistics" ③ so a check mark appears, then select "OK" ④.



	A	B	C	D
1	Before	After	Difference	
2	142	140	-2	
3	140	136	-4	
4	144	147	3	
5	144	139	-5	
6	142	143	1	
7	146	142	-4	
8	149	145	-4	
9	148	145	-3	
10	142	138	-4	
11	148	146	-2	
12				
13				
14				

Excel will now show you the summary statistics.
we have added highlights to show the mean, standard deviation, standard error, and sample size which are the summary statistics that you will most commonly use for your reports.



	A	B	C
1	Before	After	Difference
2	142	140	-2
3	140	136	-4
4	144	147	3
5	144	139	-5
6	142	143	1
7	146	142	-4
8	149	145	-4
9	148	145	-3
10	142	138	-4
11	148	146	-2
12			
13			
14			
15	*Column1*		
16			
17	Mean	144.5	
18	Standard Error	0.9803627	
19	Median	144	
20	Mode	142	
21	Standard Deviation	3.1001792	
22	Sample Variance	9.6111111	
23	Kurtosis	-1.4032926	
24	Skewness	0.1957746	
25	Range	9	
26	Minimum	140	
27	Maximum	149	
28	Sum	1445	
29	Count	10	

Data Analysis and Graphing in Excel

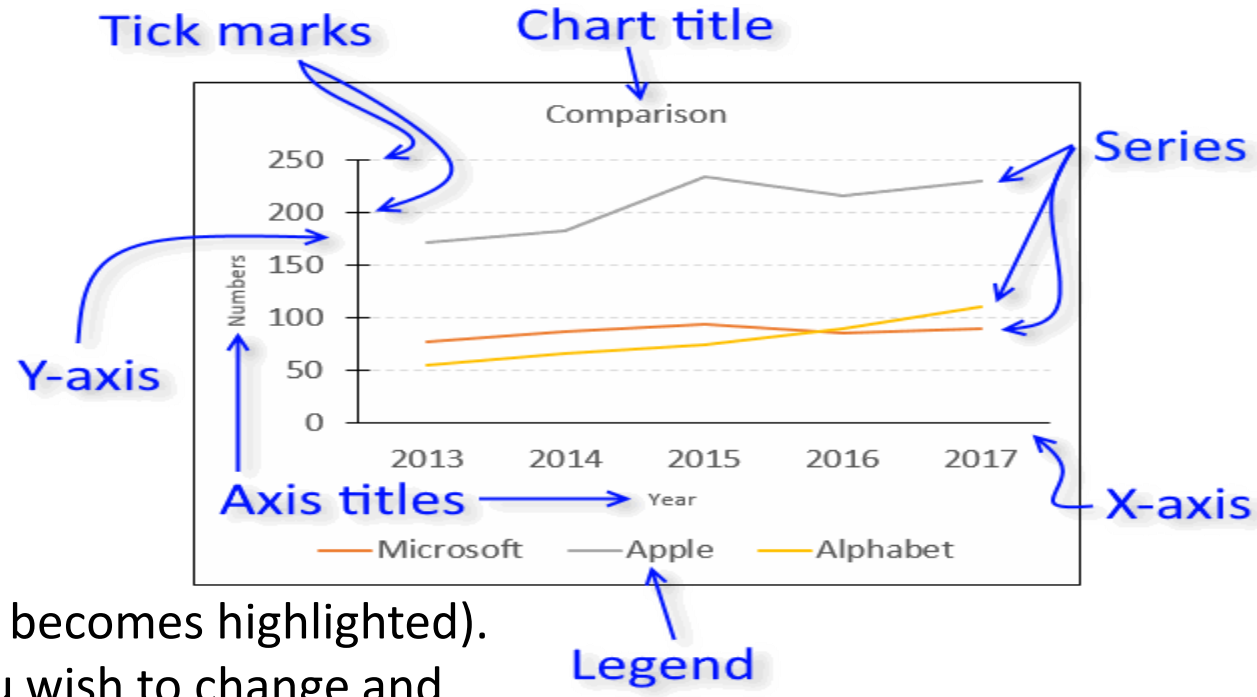
3- How to Graph Data in Excel

To create a chart:

- Select the data to appear in the chart (with labels if relevant)
- Use the Insert tab and Charts Group and click on the relevant chart in this group.

The general approach is to note that the chart has several areas:

- Chart Title
- Plot Area
- Axis titles
- The x-axis (for charts other than pie chart) and Labels
- The y-axis (for charts other than pie chart) and Labels
- Legend Entry (explains the symbols used in the chart)



To edit an existing chart

- Select the chart (click inside the chart and the border becomes highlighted).
- Then select within the chart that part of the chart you wish to change and right click to obtain the menu for reformatting the chart.

Data Analysis and Graphing in Excel

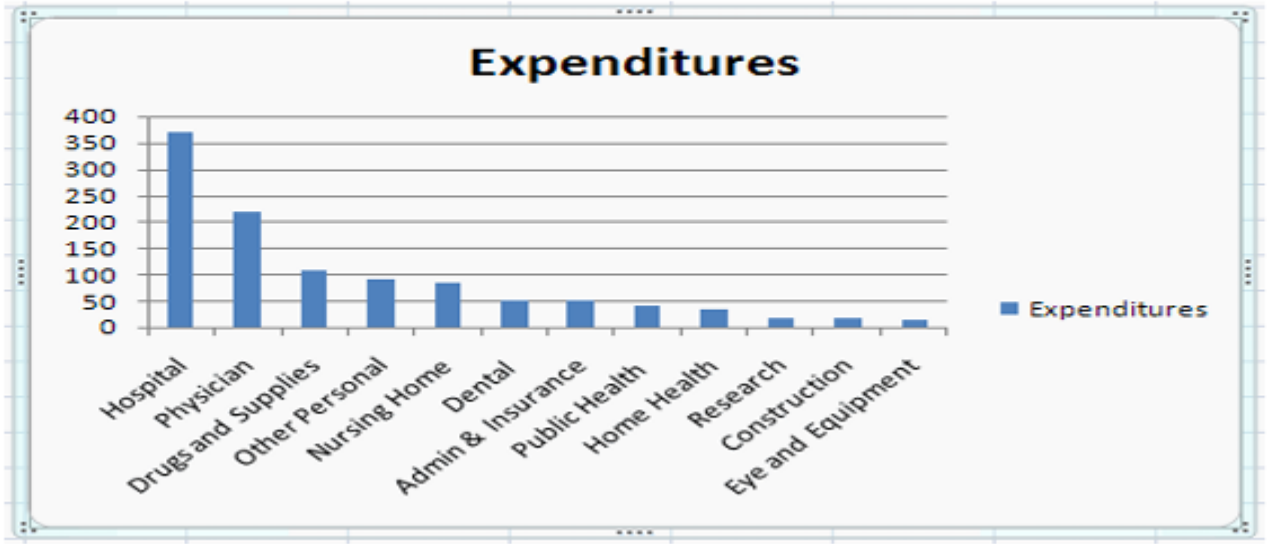
3- How to Graph Data in Excel

Bar (Column) Graph

- As an example, consider categorical data on Health Care Expenditures in billion US dollars.
- To create a Bar chart from these data
  - Highlight the data and headings (here columns A1:B13)
  - Select the Insert Tab and Charts group and click on Column Chart
  - Select the first of the 2-D Column Charts (a clustered column chart).

This yields:

	A	B
1	Category	Expenditures
2	Hospital	371
3	Physician	218
4	Drugs and Supplies	109
5	Other Personal	92
6	Nursing Home	83
7	Dental	51
8	Admin & Insurance	50
9	Public Health	39
10	Home Health	32
11	Research	18
12	Construction	17
13	Eye and Equipment	14



Data Analysis and Graphing in Excel

3- How to Graph Data in Excel

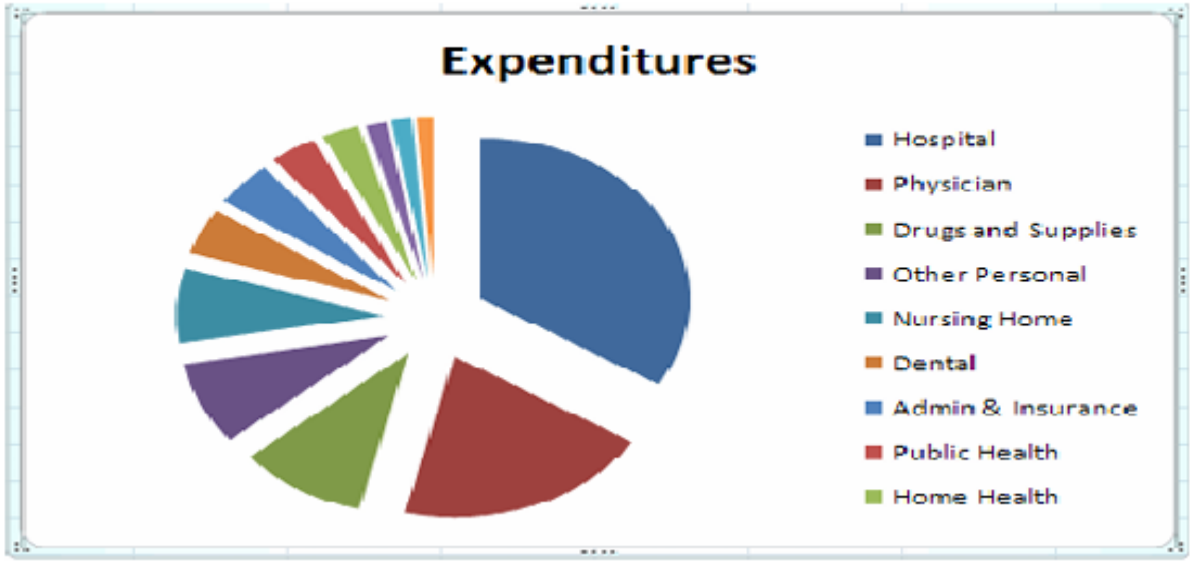
Pie Chart

- To create a pie chart

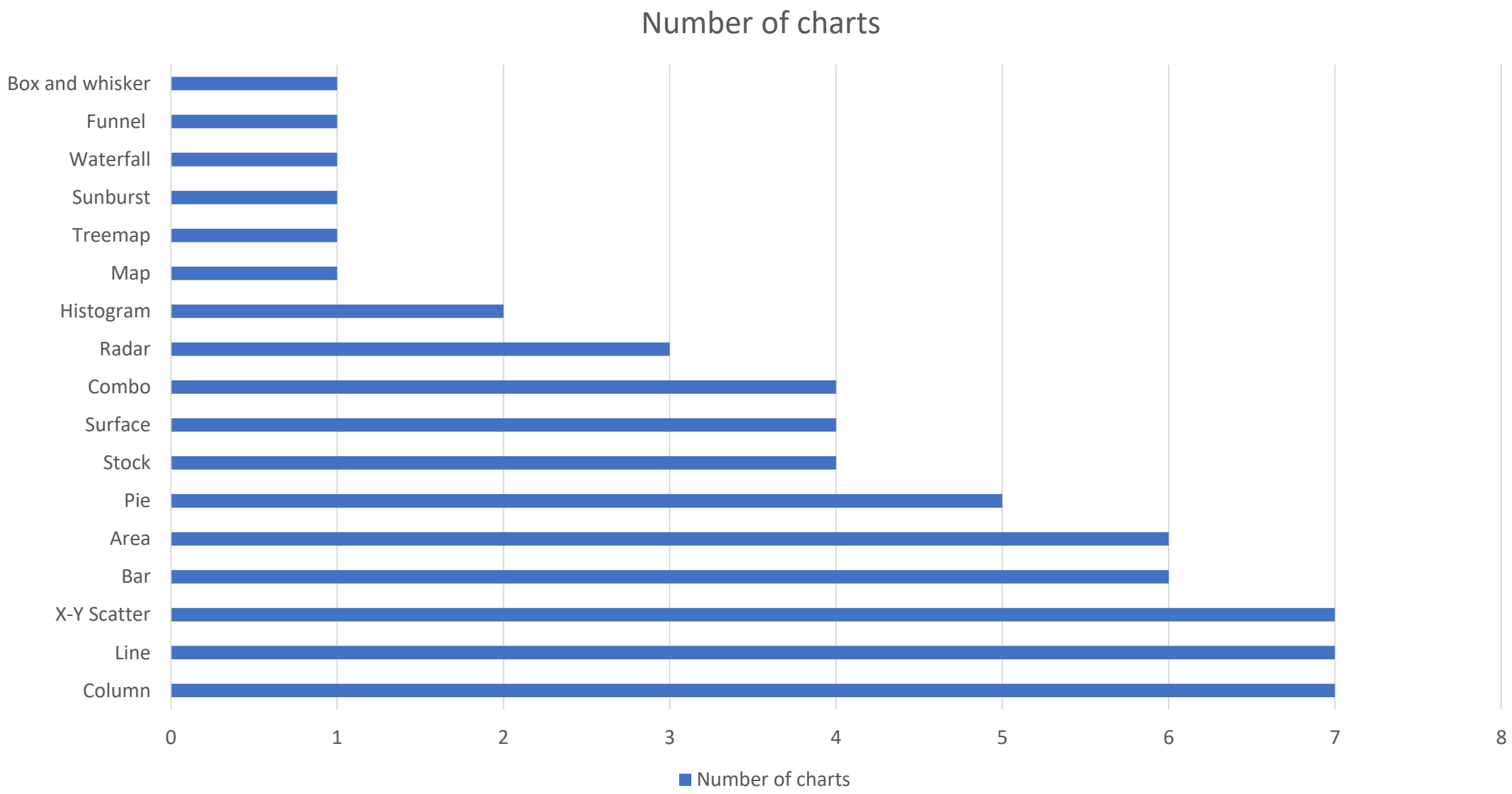
- Highlight the data and headings (here columns A1:B13)
- Select the Insert Tab and Charts group and click on Pie Chart
- Select the first of the 2-D Pie Charts (an exploded pie chart).

This yields:

	A	B
1	Category	Expenditures
2	Hospital	371
3	Physician	218
4	Drugs and Supplies	109
5	Other Personal	92
6	Nursing Home	83
7	Dental	51
8	Admin & Insurance	50
9	Public Health	39
10	Home Health	32
11	Research	18
12	Construction	17
13	Eye and Equipment	14



About 60 types of charts



Assignment

➢ Prepare a formal technical report following the basic structure of a technical report.

- The report must be prepared using **Microsoft Word** for document design and formatting, and **Microsoft Excel** for data representation and the creation of graphs and charts.
- Select any topic of personal interest, provided that it is suitable for a formal technical report.

Submission Deadline: Tuesday, 17 November 2026.

END of LECTURE